Standard Errors and Confidence Intervals for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

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Abstract

Mediation modeling using longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive (DT-VAR) approaches are commonly used to estimate indirect effects in longitudinal data, they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. To address these issues, continuous-time vector autoregressive (CT-VAR) models have been proposed as an alternative. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the direct, indirect, and total effects, for a range of time-interval values, can be calculated using parameters estimated from CT-VAR models for causal inferential purposes. However, methods for calculating the uncertainty around the direct, indirect, and total effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results from using the delta and Monte Carlo methods to calculate standard errors and confidence intervals for the direct, indirect, and total effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed` `R` package.

Keywords: mediation, continuous-time, delta method, Monte Carlo method, R package, cTMed

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Hypothetical Examples

In this section, we use hypothetical examples within the CT-VAR framework to (1) elaborate on what we mean by direct, indirect, and total effects within this modeling framework, (2) develop the delta and Monte Carlo methods for generating standard errors and confidence intervals for the effects previously defined, and (3) introduce regions of significance using the delta and Monte Carlo methods to visualize statistically significant results across a range of time-intervals.

Hypothetical Example 1: Understanding Direct, Indirect, and Total Mediation Effects in CT-VAR Models

In both cross-sectional and autoregressive models, the fundamental concept of mediation analysis involves decomposing the total effect into direct and indirect components. This approach allows us to understand how the mediator variable M influences the relationship between X and Y. However, in the context of autoregressive models, several authors (e.g., Aalen et al., 2016; Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) propose a conceptual shift when thinking about the lagged parameters $\boldsymbol{\beta}$. In this conceptualization, $\boldsymbol{\beta}$ is treated as the matrix of total effects rather than direct effects. This shift has implications for thinking about mediation models in VAR models particularly when it comes to decomposing the total effect into direct and indirect effects. In the following hypothetical example, we use simulated data from descriptive statistics reported by Grundy et al. (2007) to demonstrate how the direct and indirect effects are decomposed from the total effect. We used the `mvrnorm` function from the `MASS R` package (Venables & Ripley, 2002) to generate multivariate normal data using the correlation matrix reported in Table 1 of Grundy et al. (2007). The researchers tested the hypothesis that maternal knowledge mediated the association between mother reported marital conflict and child reported behavioral competence. Furthermore, they added a reciprocal

effect from behavioral competence to knowledge. The original data set was collected using a discrete time-interval of a year over a period of four years. Note that Deboeck and Preacher (2015) also use the descriptive statistics from Grundy et al. (2007) to demonstrate continuous-time mediation. While our approach is similar to that of Deboeck and Preacher (2015), for illustrative purposes, the model we fitted has a number of differences. First, we fitted a CT-VAR model using the simulated data set where both the measurement and the dynamic structures are estimated. Second, we specified all possible paths from one variable to another to be estimated in the drift matrix $\mathbf{\Phi}$. Third, we increased the sample size from 133 to 500 to increase the statistical power to detect significant effects. Prior to fitting the model in the `dynr` `R` package (Ou et al., 2019), we inserted missing values in between measurement occasions discrete measurement occasions using .10 increments to aid in the optimization process.

We label the independent variable marital conflict X, the mediator variable maternal knowledge M, and dependent variable behavioral competence Y. Since, we are incorporating a measurement model, the latent variables η_X, η_M, and η_Y represent the latent variables in the model with X, M, and Y as observed indicators. The estimated CT-VAR drift matrix $\mathbf{\Phi}$ is presented in Table 1. In the study, time t is scaled in years and $\Delta t = 1$ a time-interval of one year between measurement occasions given by Figure 1a. Given Equation **??** we can calculate the matrix of lagged coefficients for a time-interval $\Delta t = 1$ from the drift matrix $\mathbf{\Phi}$ as follows:

$$
\begin{aligned}
\boldsymbol{\beta}_{\Delta t(1)} &= \exp\left(\mathbf{\Phi}\right) \\
&= \begin{array}{c} \\ \eta_{X_{t+1}} \\ \eta_{M_{t+1}} \\ \eta_{Y_{t+1}} \end{array}
\begin{array}{c} \begin{array}{ccc} \eta_{X_t} & \eta_{M_t} & \eta_{X_t} \end{array} \\ \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \beta_{1,3} \\ \beta_{2,1} & \beta_{2,2} & \beta_{2,3} \\ \beta_{3,1} & \beta_{3,2} & \beta_{3,3} \end{pmatrix} \end{array}.
\end{aligned}
\tag{1}
$$

Now we look at the case of a DT-VAR model from the same continuous process except that the time-interval used is three years ($\Delta t = 3$) between measurement occasions. From the time-series literature (e.g., Hamilton, 1994) we can derive $\boldsymbol{\beta}_{\Delta t(3)}$ from $\boldsymbol{\beta}_{\Delta t(1)}$ using matrix powers. The key is to raise the matrix with the smaller time-interval by the factor in which the time-interval increased to the larger time-interval. For example, the change from $\Delta t = 1$ to $\Delta t = 3$ is a factor of three, thus $\boldsymbol{\beta}^3_{\Delta t(1)} = \boldsymbol{\beta}_{\Delta t(3)}$. Therefore, we can say that $\boldsymbol{\beta}_{\Delta t(3)} = \exp(3\boldsymbol{\Phi}) = \boldsymbol{\beta}^3_{\Delta t(1)}$. Let us explore the relationship of the elements of $\boldsymbol{\beta}_{\Delta t(3)}$ as a function of the elements of $\boldsymbol{\beta}_{\Delta t(1)}$. In particular, we will focus on the third row and first column in $\boldsymbol{\beta}_{\Delta t(3)}$ pertaining to the association of marital conflict (η_X) at time t and behavioral competence (η_Y) at time $t + 3$. The purple line in Panel 1a in Figure 1 represents the total effect of marital conflict (η_X) at time t on behavioral competence (η_Y) at time $t + 3$. This total effect is decomposed as a function of the lagged parameters in $\boldsymbol{\beta}_{\Delta t(1)}$. By raising Equation 1 to the third power and subsetting the third row and the first column of the resulting matrix we get

$$\begin{aligned}
\text{Total} &= \boldsymbol{\beta}_{\Delta t(3)}[3,1] \\
&= \beta_{1,1}\beta_{1,1}\beta_{3,1} + \beta_{2,1}\beta_{1,2}\beta_{3,1} + \beta_{3,1}\beta_{1,3}\beta_{3,1} \\
&+ \beta_{1,1}\beta_{2,1}\beta_{3,2} + \beta_{2,1}\beta_{2,2}\beta_{3,2} + \beta_{3,1}\beta_{2,3}\beta_{3,2} \\
&+ \beta_{1,1}\beta_{3,1}\beta_{3,3} + \beta_{2,1}\beta_{3,2}\beta_{3,3} + \beta_{3,1}\beta_{3,3}\beta_{3,3},
\end{aligned} \tag{2}$$

where each term represents distinct paths from marital conflict (η_X) at time t to behavioral competence (η_Y) at time $t + 3$ graphically represented by Panels 1b to 1j in Figure 1. Note that red paths in are paths where maternal knowledge (η_M) is traversed. We define these red paths as indirect effects of marital conflict (η_X) at time t on behavioral competence (η_Y) at time $t + 3$ through maternal knowledge (η_M). The sum of these red paths constitute the overall indirect effect. The blue paths are the direct effects and their sum constitutes the overall direct effect of marital conflict (η_X) at time t on behavioral

competence (η_Y) at time $t+3$. The sum of the direct and indirect effects represents the total effect of marital conflict (η_X) at time t on behavioral competence (η_Y) at time $t+3$.

Now we divide any arbitrary time-interval Δt into infinitely many smaller parts and insert these infinitely many latent variables between values at time t and time $t+\Delta t$. The idea in the CT-VAR approach is that the lagged parameters from the discrete-time integral solution represents the sum of all the effects from the infinitely many smaller parts in between time t and time $t+\Delta t$. The decomposition we discussed earlier still apply. How do we then obtain the direct, indirect, and total effects? The total effect is straight forward as it is simply the result of exponentiating the product of the drift matrix and the time-interval. Using the estimated drift matrix presented in Table 1 and a time-interval $\Delta t = 3$, the total effect matrix is given by

$$\begin{aligned}
\text{Total} &= \exp\left(\Delta t \boldsymbol{\Phi}\right) \\
&= \exp\left[3\begin{pmatrix} -0.034 & 0.038 & -0.013 \\ -0.152 & -0.586 & 0.228 \\ -0.100 & 0.104 & -0.512 \end{pmatrix}\right] \\
&= \begin{pmatrix} 0.892 & 0.049 & -0.006 \\ -0.240 & 0.183 & 0.139 \\ -0.172 & 0.056 & 0.238 \end{pmatrix}.
\end{aligned} \tag{3}$$

Following Ryan and Hamaker (2021), the matrix of direct effects is obtained by removing the effects in the drift matrix $\boldsymbol{\Phi}$ that originates from and/or goes to the mediator (or mediators) before exponentiating. We do this by pre- and post-multiplying the drift matrix $\boldsymbol{\Phi}$ with a diagonal matrix $\mathbf{D_m}$ where the ith and j^{th} index pertaining to the mediators are set to zero. In our current example where maternal knowledge (η_M) is treated as the mediator variable, the direct effect matrix is given by

$$
\begin{aligned}
\text{Direct} &= \exp\left(\Delta t \boldsymbol{D_m} \boldsymbol{\Phi} \boldsymbol{D_m}\right) \\
&= \exp\left[3 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -0.034 & 0.038 & -0.013 \\ -0.152 & -0.586 & 0.228 \\ -0.100 & 0.104 & -0.512 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right] \\
&= \begin{pmatrix} 0.908 & 0.000 & -0.019 \\ 0.000 & 1.000 & 0.000 \\ -0.145 & 0.000 & 0.218 \end{pmatrix}.
\end{aligned} \tag{4}
$$

The indirect effect can now be obtained by subtracting the direct effect matrix from the total effect matrix.

$$
\begin{aligned}
\text{Indirect} &= \text{Total} - \text{Direct} \\
&= \exp\left(\Delta t \boldsymbol{\Phi}\right) - \exp\left(\Delta t \boldsymbol{D_m} \boldsymbol{\Phi} \boldsymbol{D_m}\right) \\
&= \begin{pmatrix} -0.016 & 0.049 & 0.013 \\ -0.240 & -0.817 & 0.139 \\ -0.027 & 0.056 & 0.021 \end{pmatrix}
\end{aligned} \tag{5}
$$

We subset the ith row and j^{th} column of the resulting matrices for the direct, indirect, and total effects of the mediator variable/s on the path from the jth column variable (1 for marital conflict in our case) to the ith row variable (3 for behavioral competence in our case). The direct, indirect, and total effects of marital conflict (η_X) on behavioral competence (η_Y) through maternal knowledge (η_M), for a time-interval of 3, are -0.145, -0.027, and -0.172, respectively.

The `Med` function in the `cTMed` `R` package calculates the direct, indirect, and total effects for a specific time-interval or a range of time-intervals. A plot method for the `Med`

function can be used to visualize the effects as a function of time-interval.

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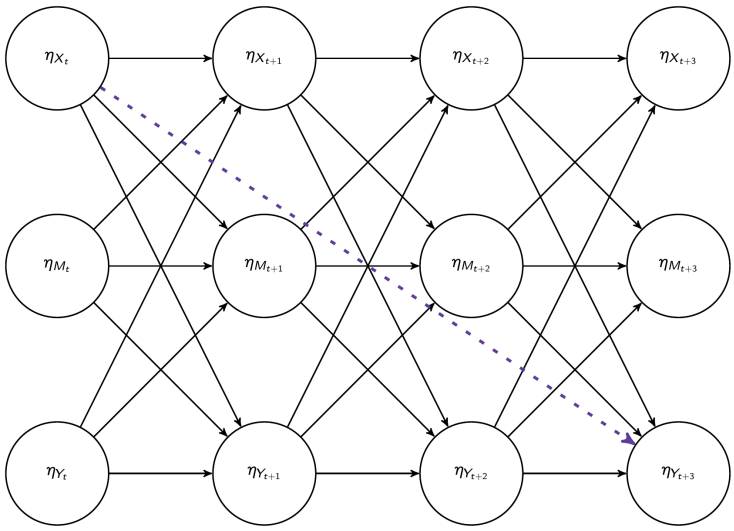
Table 1

Point and Interval Estimates of the CT-VAR Drift Matrix $\mathbf{\Phi}$ *for Data Simulated from Grundy et al. (2007)*

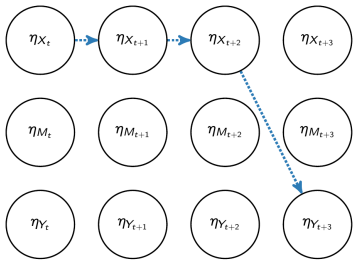
Parameter	Estimate	Standard Error	2.5%	97.5%
Conflict to conflict ($X \to X$)	-0.034	0.012	-0.056	-0.011
Conflict to knowledge ($X \to M$)	-0.152	0.031	-0.213	-0.090
Conflict to competence ($X \to Y$)	-0.100	0.029	-0.157	-0.043
Knowledge to conflict ($M \to X$)	0.038	0.019	0.000	0.075
Knowledge to knowledge ($M \to M$)	-0.586	0.091	-0.764	-0.408
Knowledge to competence ($M \to Y$)	0.104	0.055	-0.004	0.212
Competence to conflict ($Y \to X$)	-0.013	0.017	-0.047	0.020
Competence to knowledge ($Y \to M$)	0.228	0.059	0.113	0.343
Competence to competence ($Y \to Y$)	-0.512	0.075	-0.659	-0.364

Figure 1

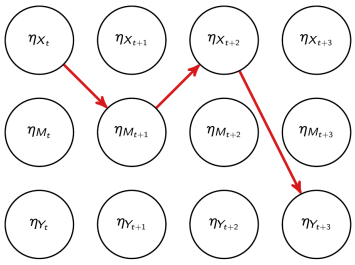
Decomposing the Direct and Indirect Effects from the Total Effect (Bidirectional)



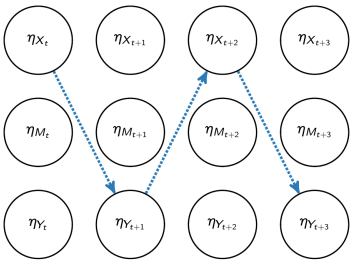
(a) *DT-VAR with time-interval of one. The purple line represents the total effect of X on Y at a time-interval of three.*



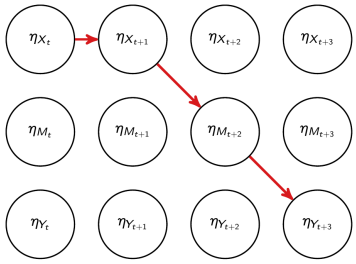
(b) $\beta_{1,1}\beta_{1,1}\beta_{3,1}$



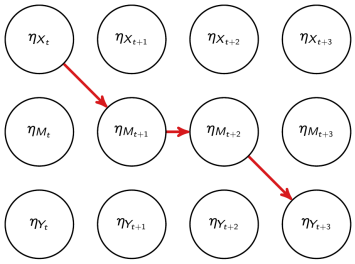
(c) $\beta_{2,1}\beta_{1,2}\beta_{3,1}$



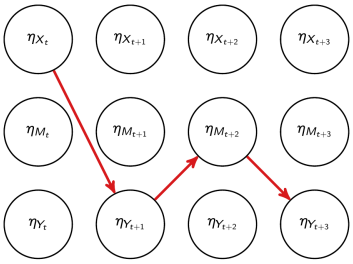
(d) $\beta_{3,1}\beta_{1,3}\beta_{3,1}$



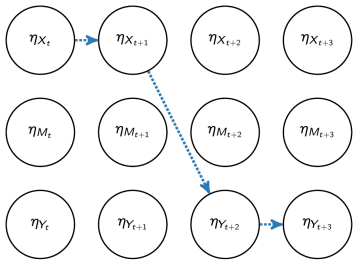
(e) $\beta_{1,1}\beta_{2,1}\beta_{3,2}$



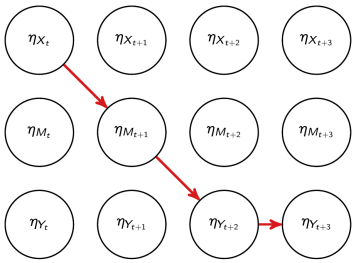
(f) $\beta_{2,1}\beta_{2,2}\beta_{3,2}$



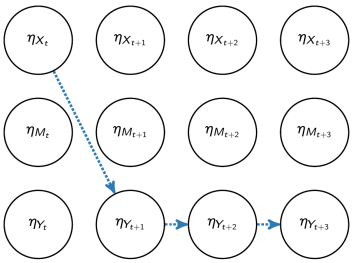
(g) $\beta_{3,1}\beta_{2,3}\beta_{3,2}$



(h) $\beta_{1,1}\beta_{3,1}\beta_{3,3}$



(i) $\beta_{2,1}\beta_{3,2}\beta_{3,3}$



(j) $\beta_{3,1}\beta_{3,3}\beta_{3,3}$

Note: Panel (1a) represents the total effect of η_X on η_Y at a time-interval of three is a function of the lagged parameters for a time-interval of one. Panels (1b) to (1j) represent the decomposition of the total effect (purple) into direct (blue) and indirect (red) effects.